

One Timestamp Does Not Identify Temporal Support in Event-Detection Benchmarks

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Abstract

Event-detection benchmarks compare a prediction with a label at a single nominal timestamp, and that timestamp does not identify when the evidence supporting the prediction was acquired or which instant its output represents. This paper separates the two quantities on released artifacts. For a released recurrent detector, occluding each bin of the newest 50 ms input window puts that window's influence centroid 23.81 ms before the label instant, and the recurrent state extends the evidence further back: occluding whole past windows leaves the newest with 22.7 % of the influence. A regression of its detections on the label velocity, carrying a placebo term no temporal offset can reach, places the effective output at -2.40 ± 7.18 ms, consistent with the label instant and 3.65 standard errors from the newest-window centroid; across specifications with geometric controls, none places the output at the newest-window centroid. Average precision is nearly invariant over this measured separation: displacing the ground truth across the interval moves mAP by 0.04 points and reorders none of 5 released checkpoints, while the interval is 0.61 px of displacement at the ninetieth percentile of label speed, the size of the labels' own annotation dispersion, and 1.29 px at the ninety-ninth. An unchanged score therefore does not establish temporal alignment; the metric is insensitive over states the predictor-side measurements distinguish. The same ambiguity appears on the label side: DSEC's six ceiling-exposure sequences carry the harmonic series of a 100.00 Hz mains flicker, undocumented in the release, and event timing inside one such exposure cannot be attributed to object timing until that illumination-driven structure is separated.

1 Introduction

An object detection benchmark evaluates a prediction $\hat{s}$ against a ground truth s^* through a function $\rho(\hat{s}, s^*)$ of position, shape and class, indexed by a query time t . The query time selects which label the prediction is compared against. The metric itself has no argument in which a time could appear, so a temporal discrepancy is scored as displacement. Neither side of the comparison is a state at an instant. A predictor computes $\hat{s}$ from a window of sen-

sor data. A label is produced by an exposure, an annotator or tracker working on the exposed image, and a geometric transfer into another sensor's frame.

The predictor side of that mismatch is fixed by a released configuration and is measurable from released artifacts. The event branch of the detector examined here [23] consumes a window of 50 ms divided into ten bins, and the preprocessing script of its own repository builds the window boundaries by counting backwards from label timestamps, so each window ends at the label it is scored against (Sec. 3.2). Under uniform weighting the evidence in such a window is centered 25 ms before the instant the label asserts. Whether the trained network weights its bins uniformly is a separate question, and occluding each bin in turn on real validation samples answers it: the influence-weighted centroid is -23.81 ms (Sec. 3.3). A network whose newest window sits there and whose output is scored at the label instant extrapolates over that interval.

The training objective already places the output at the label instant, and the measurement of Sec. 3.4 is consistent with it, at -2.40 ± 7.18 ms, which is a failure to reject and not a demonstration that they coincide. The objective does not fix the interval that measurement leaves open, 26.2 ms between that window's centroid and the measured output time. The velocity that measurement regresses on is finite differenced from the same labels the residual is measured against, and a forward difference shares their noise, returning an offset from a detector that has none. The estimator is therefore calibrated on synthetic tracks before it is used, and carries a placebo term on which no lag can load.

The second measurement concerns the same class of undeclared timing property on the label side. DSEC publishes each frame's own exposure interval, and across the eighteen training sequences whose exposure files resolve, the width varies by a factor of 127 against a frame period fixed at 50 ms, with 32.5 % of frames pinned at an auto-exposure ceiling of $14\,996\,\mu\text{s}$ in six sequences (Fig. 1). Those six carry the harmonic series of a 100.00 Hz mains flicker, and between 95.4 % and 100.0 % of labeled boxes carry more power at the fundamental than at an off-frequency arm on the same events (Sec. 4). A $10\,000\,\mu\text{s}$ period inside a $14\,996\,\mu\text{s}$ exposure displaces event times with no motion involved, and Sec. 4.4 reports what that leaves of a per-object statistic.

The single timestamp the metric is indexed by does not

specify the temporal support of either side of the comparison. Three quantities are collapsed into it, and this work separates and measures each on released artifacts: the evidence support of a released recurrent event detector, whose newest window is centered 23.81 ms before the label instant and carries 22.7 % of its influence, the rest reaching past the second measured (Secs. 3.2–3.3); the output time of the same checkpoint, identified by a vector regression with a placebo term and a sign-flip control on the velocity estimator, which places it at the label instant and 26.2 ms from that window (Secs. 3.4–3.5); and the exposure-conditioned support of DSEC’s labels, where the harmonic series of a 100.00 Hz mains flicker in six ceiling-exposure sequences bounds what a per-object temporal statistic measured inside one published exposure window can be attributed to (Sec. 4).

The consequence claimed is a failure of identifiability under the metric, not a ranking change. Displacing the ground truth across the measured interval moves mAP by 0.04 points and reorders none of the 5 released checkpoints, while the same interval is 0.61 px of displacement at the ninetieth percentile of label speed and 1.29 px at the ninety-ninth; Tab. 2 locates the insensitivity in the speed composition of the benchmark rather than in an absence of displacement.

This work makes three contributions. First, we separate evidence support from effective output time in released event detectors and show that the two are measurably distinct. Second, we show that standard average precision is nearly invariant over that measured separation, so benchmark agreement does not establish temporal alignment. Third, we identify a complementary label-side limit in DSEC, where exposure-conditioned temporal structure prevents within-exposure event timing from being uniquely attributed to object timing.

2 Related work

Recovering a time offset from image-plane displacement. Qin and Shen [5] calibrate a clock offset online by modeling a feature observation as $z(t_d) = [u, v]^T + t_d V$, with V differenced from consecutive observations. That is the estimator of Sec. 3.4, published in visual-inertial odometry, and the same work identifies the offset by making one scalar reconcile two channels. Temporal calibration removes the offset between two *sensor* clocks by pooling over features, so the spread of per-feature estimates is its error bar. Here the offset lies between a predictor’s clock and a *benchmark’s* label clock. Streaming perception [6] folds wall-clock compute latency into average precision and re-anchors any detector from box coordinates alone; later work adapts to the measured run-time delay [35], a continuous-stream framework scores

the latest available output against high-rate ground truth and reports reordered rankings [29], and temporal tolerance has been argued for time-series event detection [36]. Each of those latencies is the time from an event to an output and falls as the device gets faster. Sensor latency, the illumination-dependent delay between a change and the timestamp the sensor assigns it, is measured and corrected at microseconds [28], and nighttime event imaging corrects an illumination-dependent trailing by learning a per-event timestamp calibration [24]; both target the sensor, while here the released exposure metadata and event stream bound what a per-object temporal statistic computed inside one published exposure can be attributed to. Inside the citation graph of LET-3D-AP [1], a latency-aware average precision scores 3D detection at the instant a prediction becomes available on a named device [37], again compute latency; the interval measured in Sec. 3.5 is fixed by a data representation. The event–RGB frequency mismatch is named in [38], which scores with unmodified average precision.

Decomposing localization error along a privileged axis.

LET-3D-AP [1] decomposes camera-only 3D localization error along the line of sight and defines an average precision tolerant of the longitudinal part, and reports a camera-to-label synchronization gap as a per-dataset interval that depends on horizontal object position. Sec. 3.4 keeps the idea of a privileged axis but replaces the sensor’s line of sight with the object’s own velocity vector, and estimates a time rather than defining a tolerant score. TIDE [7, 8] and DETAD [9] bucket detection error atemporally, so a temporal offset falls inside their localization bucket, HOTA [10] takes the label clock as exact, and the response of average precision to isotropic box perturbation [11] supplies the isotropic null the supplement uses.

Per-object time in labels. Waymo’s `camera_synced_box` solves t_{capture} per object and stores the resulting box shift rather than the time [2, 3]. Nuñez et al. [4] measure within-frame per-object annotation offsets on Argoverse 2 and TruckScenes and define a dispersion of the *spatial* error. Each computes a per-object time in order to remove it. The event-vision label formats carry the field: the Prophesee dtype used by Gen1, the 1 Mpx dataset and DSEC-Det has a per-box τ . What no release fills is that field’s per-object degree of freedom. Across the 60 released DSEC-Det sequences, 390 118 boxes take 70 379 distinct values of τ , 5.54 boxes per value, and those values are the 50 ms frame clock.

Event benchmarks and their exposures. Gen1 [16] labels are drawn on gray-level images assembled from asynchronous per-pixel measurements. On the 1 Mpx

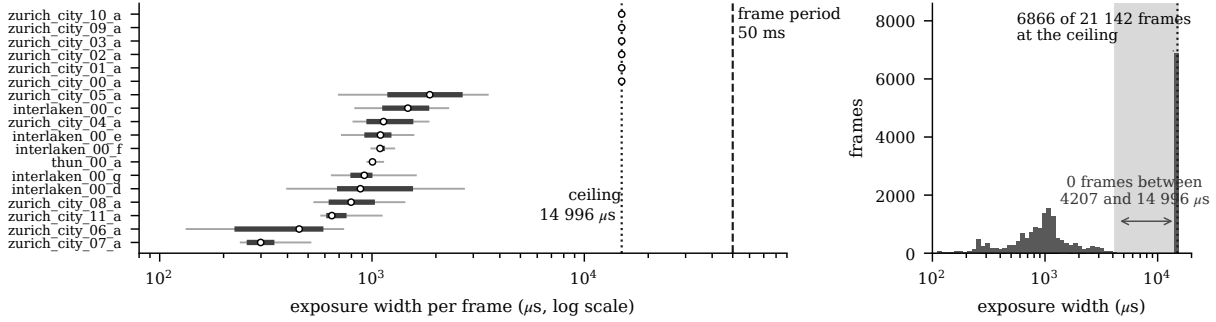


Figure 1: Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve: per sequence (left, log axis, median, interquartile range and 5–95 % band, against the fixed 50 ms frame period) and pooled over 21 142 frames (right). 32.5 % of frames sit at the auto-exposure ceiling of 14 996 μ s and zero fall between the largest non-ceiling width of 4207 μ s and that ceiling, so the width is a two-level factor on this data; the six ceiling sequences are the ones measured in Sec. 4.

dataset [17] the event count inside a labeled box at its own timestamp is often too small to support detection [18]. DSEC-Det [13] labels by tracking [14] on rectified frames and warps into the event view, and Ev-3DOD [19] interpolates lower-rate annotations into high-rate 3D labels. That a frame’s exposure is long relative to event timing has been stated: REFID [20] argues that accumulating events from a single frame timestamp loses information, deblurring work names auto-exposure as why the width is unknown [21], and RENet [22] uses a DSEC exposure interval as its event window while describing it as short; none reads the released per-frame widths or measures what is inside one. State space backbones [25] removed by architecture the dependence of an event model’s score on inference-time window duration, and adaptive spatial-temporal windows [27] retrain a detector under each of nine partitioning strategies and report the score of each. Both treat the window as a design variable to choose. Here it is a released fact to locate. That the released window is causal is already in print: LEOD [26] states that event detectors take only events triggered before the query time, and builds a time-flip augmentation on the consequence that forward inference favors receding objects; where inside that window the evidence sits is not measured there. Protocol work supplies the standards these measurements are held to [31–34]: a repaired protocol can reverse a field’s conclusion, and a reported score is an estimate with a variance. A protocol audit can expose failures in what a benchmark identifies even when the resulting correction reorders no existing method; a ranking change establishes one form of consequence, and failure to distinguish two measurably different states establishes another. The claim here is of the latter type: average precision is nearly invariant over a temporal interval that the released predictor’s evidence and output measurements distinguish.

3 The interval between evidence and label

3.1 Definition

Let a predictor P emit a state at query time t with normalized temporal support kernel $k_P(u)$, the share of the emitted state’s information coming from $t + u$, of centroid c_P . A benchmark scoring at a nominal instant treats that centroid as zero. Of the benchmarks and released detectors whose documentation we read, Gen1, 1 Mpx, DSEC-Det, DSEC-3DOD, DSEC-Flow, MVSEC, the RVT release and the SSM-ViT release [25], none states or checks it.

The *effective timestamp* τ_P is not c_P . For a predictor linear in position measurements at bin centers c_k with weights a_k summing to one, the conditional expectation of the output under constant velocity is the true position at $\tau_P = \sum_k a_k c_k$. Non-negative weights give $\tau_P = c_P$, while a predictor that uses older evidence to compensate its own lag has negative weights and moves τ_P toward zero without moving c_P . Sections 3.2 and 3.3 measure the newest window’s contribution to c_P , Sec. 3.4 measures τ_P , and Sec. 3.5 reports the interval between them.

3.2 The window and the instant it is indexed by

Two released files fix the input side. The dataset configuration `config/dataset/gen1.yaml` of the detector’s own repository [23] names the representation `stacked_histogram_dt=50_nbins=10`, a window of 50 ms divided into ten bins of 5 ms. Line 405 of `scripts/genx/preprocess_dataset.py` constructs the array `ev_repr_timestamps.us_end` by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive label timestamps. Each representation is therefore indexed

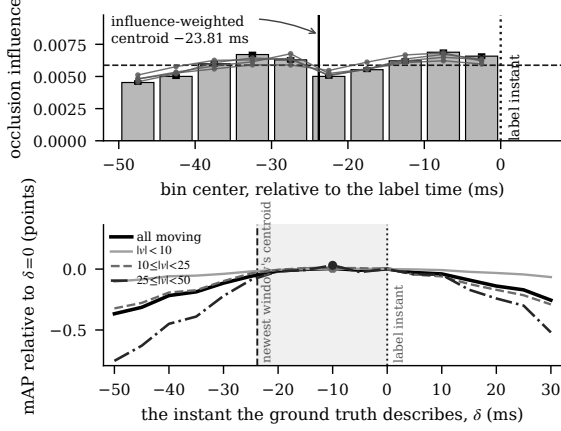


Figure 2: The detector’s evidence and output are temporally distinguishable, while average precision is nearly invariant over the interval between them. (a) Occlusion influence of each of the ten bins of the released `rvt-t` input window, over 480 real Gen1 validation samples with the recurrent state warmed eight steps, with standard errors. The dashed line is the mean over bins, the dotted line the label instant, and the solid line the influence-weighted centroid; grey lines are the 4 other checkpoints, each rescaled to its own mean. (b) Average precision against the instant δ the ground truth is displaced to describe, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum marked. The shaded band is the interval between the newest window’s influence centroid and the label instant; its width is 0.61 px at the ninetyth percentile of label speed and 1.29 px at the ninety-ninth, against an annotation dispersion of 0.6353 px (Tab. 2). The curves deepen with speed, and all of them peak between the two lines.

by the end of its window, and those ends are placed on label times. A window attached to a label at time t holds the events of $[t - 50 \text{ ms}, t]$, whose uniform weighting centroid falls 25 ms before the instant t at which the label asserts a state. Both statements are read from the released source, and the 25 ms figure is a reference point rather than a property of the trained network.

3.3 The measured influence profile

τ_P is defined on influence weights, which are measurable for a released checkpoint. We take 480 samples of the released Gen1 validation split with the released `rvt-t` checkpoint, warm the recurrent state for eight steps, and occlude each bin in turn, reading a bin’s influence as the relative L_2 change its occlusion produces in the emitted detection tensor. The recurrent state given to the occluded pass is the state the unoccluded one received, and the state it returns is discarded, so the profile is a property of the current window and not of a perturbed history.

The profile varies across bins and is balanced between halves (Fig. 2a): the coefficient of variation across the ten bins is 0.131, the heaviest bin carries $1.52\times$ the lightest, and the five older bins hold 48.6 % of the influence against 51.4 % for the five newer. The influence-weighted centroid is therefore -23.81 ms against the uniform-weight value of

-25.00 ms, a difference of $+1.19$ ms or 23.8 % of one bin width. For this configuration the centroid of the input is the uniform-weight centroid to within a fraction of one bin, so the configuration fact of Sec. 3.2 is a measured property of the trained network. This is the current window’s contribution to k_P , taken with the recurrent state held at the value entering the step, and not k_P itself. Zeroing a bin also pushes the input off the data manifold, so the fill is varied and the instrument replaced. Setting a bin to its dataset mean gives -26.35 ms and taking it from an unrelated sample -25.01 ms; a gradient sensitivity, which occludes nothing at all, gives -22.35 ms. The four centroids span 4.01 ms against a bootstrap standard error of 0.054 ms on the zero-fill centroid, so the spread is the choice of instrument.

The rest of the kernel. The detector is recurrent, so k_P does not end at the window. Restoring the state saved before the step at $t - 50L$, occluding that whole window and letting the state propagate to t measures each past window’s influence over 576 samples. The newest carries 22.7 % of the total over a 1000 ms horizon, half is reached only by -175 ms, and zeroing the state changes the output by 0.0867 of its norm against 0.0231 for removing the newest window entirely. The tail does not decay to zero within that second, so the influence-weighted centroid follows the horizon instead of converging: -161 ms over 500 ms of history and -299 ms over 1000 ms. Filling a past window from an unrelated sample rather than with zeros deepens the plateau, the decay across the second falling from 10.3 to 2.2. This predictor’s support therefore has no centroid the measured horizon identifies.

Across the released checkpoints and a second architecture. RVT releases three Gen1 checkpoints differing only in embedding width, attention head dimension and FPN depth. SSM-ViT [25] releases two more in which each stage’s ConvLSTM is replaced by an S5 state space layer, at 9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54; it distributes RVT’s own preprocessed Gen1 and inherits its window construction. On the same sequences, and driving each the way its own release drives it, all 5 checkpoints put the newest window’s centroid between -23.76 and -24.72 ms, none further than 1.24 ms, or 25 % of one bin, from the uniform-weight value. The state the SSM release carries between evaluation chunks reaches none of its outputs, which bounds that model’s support at the 21-window chunk its evaluation steps through; the supplement measures it. The support behind that window is not fixed in the same way either: over 576 samples of the three RVT capacities, the newest window’s share of a 1000 ms horizon falls from 22.7 % to 19.0 % and the support centroid moves from -299 to -339 ms.

3.4 The output time

The influence profile is a statement about the network’s input; a network that weights its window evenly can still emit a state describing a later instant, and that quantity is measured from the same checkpoint’s own detections.

A detector reporting the state an object occupied τ earlier displaces its prediction by $-\tau v^*$, where v^* is the label velocity, so $\tau = -\tau_P$ on the axis of Sec. 3. That is a statement about a vector, and it is estimated as one. With $\hat{c}$ the predicted center, c^* the label center, $\hat{u} = v^*/\|v^*\|$ and R a quarter turn,

$$\hat{c} - c^* = b - \tau v^* + \rho Rv^* + \gamma\sqrt{A}\hat{u} + \varepsilon \quad (1)$$

is fitted by stacking both components of every match into one least squares problem, with standard errors clustered by sequence. It is the temporal calibration of [5] carried to a box center, taken as a slope rather than as the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|v^*\|$, whose variance diverges as its denominator approaches zero. Fitted on these rows its mean is $+24.22 \pm 27.73$ ms, on the very hypothesis Sec. 3.5 excludes, with an interquartile range of 430 ms. Written this way τ is identified by construction under the model: no magnitude of the residual is taken, and τ does not load on a direction estimated from the same noisy velocity.

Rv^* is the label velocity rotated a quarter turn: it has the same magnitude and measurement noise as v^* , and no temporal offset can load on it, so ρ is a placebo whose null distribution the estimator supplies for itself. $\sqrt{A}\hat{u}$ absorbs a displacement along travel proportional to box side, which is needed because that residual grows with box size and box size correlates with label speed at $+0.139$ on this split.

The finite difference used for the velocity. v^* is finite differenced from labels, and the choice of difference changes the answer. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 ; their covariance is $\text{Var}(\delta)/\Delta$ when τ is exactly zero. A backward difference carries δ_k with the opposite sign and produces the opposite artifact. The centered difference $(c_{k+1}^* - c_{k-1}^*)/(\Delta_f + \Delta_b)$ does not use c_k^* and is free of it. Fitting all three is therefore a control on the estimator itself, calibrated before use. On 2000 synthetic tracks whose detector has no lag whatever, with label centers corrupted at the dispersion the supplement’s difference ladder measures on Gen1 (0.6515 px), the forward estimator returns -5.64 ms and the backward one $+5.84$ ms against a closed form of -5.63 ms, they sum to $+0.20$ ms, and removing the label noise returns all three to $+0.02$ ms. The same simulation recovers an injected lag at 0.92 of its size, the attenuation an errors-in-variables treatment predicts at 0.933 for this noise.

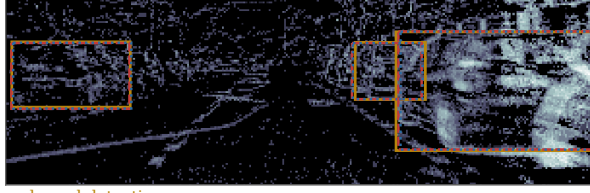
Specification	τ (ms)	placebo ρ (ms)
<i>velocity estimator, no geometric controls</i>		
forward difference	-14.68 ± 4.99	-6.53 ± 1.68
backward difference	$+45.65 \pm 5.59$	-6.15 ± 1.71
centered difference	$+17.51 \pm 5.06$	-7.76 ± 1.92
<i>controls added, centered difference throughout</i>		
+ box side along travel	$+14.62 \pm 5.47$	-7.75 ± 1.91
+ four geometric terms	-7.62 ± 7.01	-4.12 ± 1.90
+ per-sequence bias	-2.40 ± 7.18	-2.66 ± 1.71

Table 1: The output-time offset τ of the released `rvt-t` checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo ρ beside it, the coefficient on the rotated label velocity, which no temporal offset can reach. Standard errors are clustered by sequence. Under the full control set the forward and backward estimators are still 35.0 ms apart and straddle the centered one, as the shared label-noise term predicts.

The measurement. The released checkpoint is run over the Gen1 validation split with its recurrent state carried across steps, and its detections at each labeled frame are matched to the ground truth at an overlap of 0.3 or above. Requiring both label neighbors, so that a centered velocity exists, and discarding the matches whose two neighbors coincide on the integer label lattice, leaves 22 534 of 35 523 over 376 sequences at a median overlap of 0.873. Gen1’s `track_id` is a per-frame index rather than a persistent identifier, so association is rebuilt by same-class overlap between adjacent label times. No speed floor is imposed.

Table 1 is a specification ladder, and the specification was chosen by the placebo rather than by τ . Under Eq. 1 with only the box-side term, $\tau = +14.62 \pm 5.47$ ms, but ρ is -7.75 ms at 4.0 standard errors, which indicates a term omitted from Eq. 1 that Rv^* is a proxy for. A split by the sign of the horizontal velocity locates it: ρ changes sign across that split, at -27.99 ms and $+21.84$ ms, which a rotational term would not do. Most label motion here is horizontal, so Rv^* maps a horizontal speed onto a vertical residual, where a bias differing between oncoming and same-direction traffic lands with nothing rotational present. Entering box height and image row along $\hat{y}$ and box width and image column along $\hat{x}$, and then a constant vector per sequence, returns the placebo to -2.66 ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to v^* .

Under it, $\tau = -2.40 \pm 7.18$ ms, which is 0.33 standard errors from the label instant and 3.65 standard errors from the influence centroid of Sec. 3.3; a cluster bootstrap over sequences puts the 95 % interval at $[-17.0, +11.5]$ ms. The controls do not absorb a lag and the axis pairing does not make the answer: an injected offset is recovered at 1.000 of its size under the full control set, entering all four geometric terms on both axes gives -3.62 ± 7.41 ms, and across every specification carrying geometric controls τ spans -20.2 to $+11.1$ ms.



released detections
ground truth at the label instant
ground truth at the newest window's centroid, $\delta = -23.81$ ms

Figure 3: One validation frame of the sweep of Fig. 2b: released detections (solid), ground truth at its label instant (dashed) and displaced to the newest window's influence centroid (dotted). The frame is chosen by rule, the largest median label speed among frames with two labels above 20 px/s; its fastest label moves 45.2 px/s, so the two placements sit 1.08 px apart, the top of the displacement column of Tab. 2.

3.5 The interval between evidence and output

The two measurements are of one architecture at one configuration. Its newest input window is centered 23.81 ms before the label time (Sec. 3.3) and its measured output time is consistent with that label time (Sec. 3.4). The measurement does establish the separation: the output is incompatible with that window's centroid at 3.65 standard errors, and 26.2 ± 7.2 ms is the point estimate of their distance under the reported specification rather than a latency of this architecture. On the convention of Eq. 1 the centroid hypothesis is $\tau = +23.81$ ms, and no specification carrying geometric controls reaches it: τ spans -20.2 to $+11.1$ ms across them. The rest of the support lies further back, so 23.81 ms is the smallest interval consistent with the measurement rather than its value. The standard error quoted is the output-time term's; the influence centroid contributes 0.054 ms of sampling error and 4.01 ms of dependence on the instrument (Sec. 3.3), and the specification spread above is wider than either.

That outcome is the one the training objective enforces: the detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero reports the objective. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetieth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That window's centroid and the output time are both measurable from released artifacts alone.

The interval on the benchmark's own metric. The benchmark can be asked in its own units: displacing the ground truth of every labeled frame by δv^* scores the released predictions against the state each object occupied

speed (px/s)	share (%)	med. $\ v^*\ $	23.81 ms in px	Δ mAP	span
all moving	100.0	4.0	0.10	0.04	0.37
< 10	72.1	2.1	0.05	0.02	0.11
10–25	19.2	15.0	0.36	0.01	0.33
25–50	7.5	32.3	0.77	0.08	0.78
≥ 50	1.2	57.4	1.37	0.00	0.37

Table 2: The sweep of Fig. 2b by object speed, over the 27 943 ground-truth boxes carrying a centered velocity; each stratum scores its own boxes with the rest marked ignore. Δ mAP is the cost of displacing the ground truth to the newest window's influence centroid, at -23.81 ms exactly; span is the stratum's mAP range over the -50 to $+30$ ms sweep, both in points.

δ later and leaves the detector untouched; a detection carries no velocity of its own. Over 20 296 frames and 40 698 boxes, 68.7 % of which carry a centered velocity and the rest are ignored, mAP against δ is smooth with its maximum at -10 ms (Fig. 2b). On the axis of Eq. 1 that is $\tau = +10$ ms, on a top flat to 0.04 points across the interval. A bootstrap over frames puts that argmax between -20 and 0 ms in 95 % of 1000 resamples, and in none of them does it reach that window's centroid, so it identifies no value.

The flatness of that curve is the second result. The whole sweep moves mAP by 0.9 points and displacing the ground truth to that window's centroid costs 0.04 points, 0.12 % of 0.3346: the separation measured above is real and this metric does not register it. Nor does it on any of the 5 released checkpoints: displacing each to the instant the metric prefers for it leaves their ordering unchanged in four speed strata, and cannot change it, since within a stratum the largest gain from alignment is 52 % of the smallest gap between adjacent models (supplement). A displacement costs overlap in proportion to $\tau \|v^*\|$ against the box side, and the median label speed here is 4.0 px/s on a 41.5 px box, so 23.81 ms is 0.10 px; Fig. 3 shows both placements on one validation frame near the top of the speed range. Table 2 locates the flatness in that composition: 72.1 % of the moving boxes travel below 10 px/s, where the displacement is 0.05 px, and the sweep deepens to 0.78 points between 25 and 50 px/s, where it is 0.77 px, past the labels' own annotation dispersion of 0.6353 px (Sec. 3.4). Above 50 px/s the displacement reaches 1.37 px and the span shallows again, at an mAP of 0.139 against 0.286 below it: objects there are lost to detection rather than to localization, and no displacement of the ground truth returns them. The weak pooled response is a property of the benchmark's speed composition, not an absence of displacement.

4 A complementary label-side identifiability limit

4.1 Exposure and labels

The predictor side of the mismatch is fixed by a window; the label side is fixed by an exposure. DSEC publishes per-frame exposure metadata, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of $14\,996\,\mu\text{s}$ holds 32.5 % of frames, in six of the eighteen sequences, and the distribution is not continuous over that range (Fig. 1).

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences, every label sits on the frame clock, and the median spacing between consecutive label times is $50\,001\,\mu\text{s}$. Over 361 628 triples of consecutive labels on a track, 7.1 % of middle box centers fall within a hundredth of a pixel of their neighbors' linear interpolant, where interpolation would place that fraction near unity, so no construction term arises within this archive. DSEC-Det ships more than one annotation version [15]; the supplement names the one measured here.

The label center carries noise measurable on the labels themselves: on a uniform grid the variance of the k -th difference of a track's box center is $\binom{2k}{k}\sigma_c^2$ plus a motion term that halving the sampling rate multiplies by 4^k . Over 5250 DSEC-Det tracks the two routes agree from $k = 4$ upward and extrapolate to $\sigma_c = 0.4974\,\text{px}$, of which the 0.5 px integer lattice the release stores box centers on contributes 0.1443 px. Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

4.2 The line, and the per-pixel test

Swiss mains is 50 Hz, so lamp intensity flickers at 100.00 Hz; an event camera responds to log-intensity change, so both edges of that oscillation produce events and the arrival rate carries the harmonic series (Fig. 4). Sweeping 20–450 Hz on each whole recording, with no band fixed in advance and each frequency divided by the median of its own neighbourhood, returns that series in every ceiling sequence: the ceiling median is 30 at 100.00 Hz, 125 at 200 Hz and 15 at 300 Hz, and the second harmonic is the strongest line in four of the six, never falling below 14.7. At the odd multiples of 50 Hz there is nothing: 1.8, 4.5 and 1.6 at 50, 150 and 250 Hz. That absence is what a rectified supply predicts and what a periodic source of any other origin need not obey. The daytime sequence carries the fundamental at 0.5 and the second harmonic at 6.2, against 14.7 for the weakest ceiling sequence.

Per pixel, the Rayleigh statistic $Z = R^2/N$, for R the resultant length of the event phases at the test frequency,

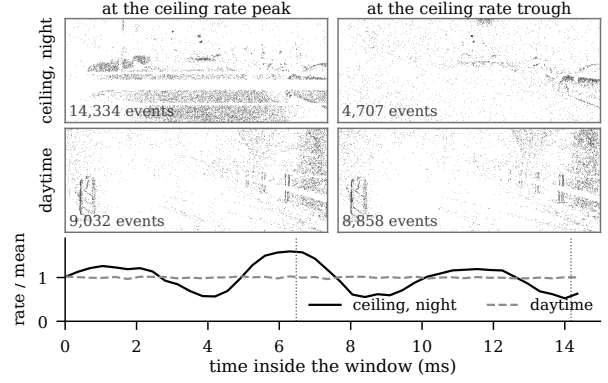


Figure 4: One $350\,\mu\text{s}$ slice of the ceiling-exposure sequence *zurich.city_09_a* and of the daytime control at the instants the ceiling sequence's own event rate peaks and bottoms inside its densest published exposure window. Bottom: that rate in $350\,\mu\text{s}$ bins, each sequence normalized to its own mean; the ceiling rate moves by a factor of 3.0 between its extremes, the daytime rate by 1.1.

is $\text{Exp}(1)$ under independent uniform phases, so its null has median 0.693 and a rate of 0.1 % at $p < 10^{-3}$. Independence is checked: over 35 990 pixels of the ceiling sequence with at least 100 events, the same estimator at 137 Hz returns 0.581 with a $p < 10^{-3}$ rate of 0.04 %, and on 17 800 daytime pixels at 100.00 Hz it returns 0.719 at 0.08 %, so per pixel the reference distribution is not exceeded off the line or away from the lamps. At 100.00 Hz on the ceiling sequence the median Z is 2.935 and 24.07 % of those live pixels reach $p < 10^{-3}$. DSEC's own description attributes high night event rates near street lamps to flashing lights [12]; neither that nor dim-light noise [30] is quantified per pixel.

4.3 All six ceiling sequences, per labeled box

The same statistic taken per labeled box, over the events inside that box during that frame's own exposure window, extends to every ceiling sequence; the supplement tabulates all six of them over 16 516 boxes. The median per-box Rayleigh statistic at 100.00 Hz runs from 68.4 to 310.9 with a median over sequences of 114.58, against 4.01 for the same estimator at 137 Hz on the same events.

The analytic null does not transfer from a pixel to a box: over eight off frequencies the pooled median is 38.4, 55 times the analytic one, since events inside one box during a $14\,996\,\mu\text{s}$ exposure arrive in bursts, so no per-box p -value is quoted, and 137 Hz is the most favorable of those frequencies, the line standing at 4.0 times their pooled median. A ratio needs no null: dividing Z at the line by the median of $Z(f)$ over each box's own neighbouring frequencies gives 1.87 over 16 516 boxes against 0.11 at 137 Hz, and 99.63 % of boxes exceed that arm's ninety-ninth percentile. One window is transform-limited

to 67 Hz, so a box establishes that the modulation reaches it and not which harmonic carries it; the identification is the sweep of Sec. 4.2.

4.4 A per-object temporal statistic under that signature

That signature bounds any per-object temporal statistic inside one published exposure window, and the bound is measurable. Let a labeled frame have published exposure $[a, b]$ of width $w = b - a$ and mid-exposure $m = (a + b)/2$, and let box i contain the events falling inside it with $t \in [a, b]$. The *event centroid* of the box, $\bar{t}_i = \frac{1}{N_i} \sum_{k \in i} t_k - m$, is the mean event time referred to the instant the dataset indexes the frame with; were the evidence inside each box uniform over $[a, b]$, $\bar{t}_i$ would have variance $w^2/12N_i$, so the excess over that null is $\sigma_{\text{evt,excess}}^2 = \text{Var}_i(\bar{t}_i) - \frac{1}{n} \sum_i w^2/12N_i$. The subtrahend holds for arrivals independent and uniform on $[a, b]$, which these are not: reassigning a frame’s box events among its boxes at random, which leaves everything common to the frame intact and destroys only box identity, gives a null of $78.6 \mu\text{s}$, 0.85 times the analytic one, and an excess of $191.1 \mu\text{s}$; the supplement gives that construction. A box enters with at least 200 events and a frame with at least 3 qualifying objects, which 1134 of 1557 labeled frames of `zurich_city_09_a` do. The within-frame standard deviation of $\bar{t}_i$ is $206.6 \mu\text{s}$ against an analytic null of $93.0 \mu\text{s}$, the excess is $183.6 \mu\text{s}$, and the standard deviation exceeds that frame’s own null on 90.1 % of frames. On `interlaken_00_c`, where the exposure is $1556 \mu\text{s}$ and the 200-event floor leaves 20 usable frames, the standard deviation is $18.0 \mu\text{s}$ against a null of $21.6 \mu\text{s}$ and the excess is zero.

The supplement tabulates what that dispersion can be attributed to. Boxes of the same shapes at random positions give $306.3 \mu\text{s}$, above the annotated boxes’, so it is a property of where evidence falls during the exposure and not of objecthood; removing the 14 334 pixels that reach $p < 10^{-3}$ leaves it at $180.6 \mu\text{s}$: a weakly modulated population remains in the clean arm. Without any mask, the excess rises monotonically with each box’s own modulation depth $m = 2|C|$ at 100.00 Hz, from 93.1 to $208.0 \mu\text{s}$ across quartiles of m over 6841 boxes, against 150.3 to $187.6 \mu\text{s}$ for the same stratification at 137 Hz, so part of the trend belongs to stratifying on any amplitude statistic, and no unmodulated stratum separates the two: m has a tenth percentile of 0.355 against a floor of 0.125. The dispersion is therefore a property of the event stream inside the published window that this data does not resolve into scene and illumination components; until they are separated, a per-object temporal statistic measured inside one of these exposures cannot be attributed to object timing.

That dispersion bounds the precision to which one

scalar timestamp denotes an instant: $183.6 \mu\text{s}$, whatever its source, displaces a box center by 0.0073 px at the median object and 0.0661 px at the ninety-ninth percentile, 1.5–13.3 % of σ_c , and the frame-common offsets are smaller, -16.8 and $-102.3 \mu\text{s}$ from mid-exposure over all events and the labeled subset. The newest window’s influence centroid of Sec. 3.3, measured on a different dataset, is larger than the dispersion by a factor of 130, and it is the one term a declared timestamp would remove.

5 Limitations

A per-object temporal claim on these sequences is bounded rather than identified: the one spatial statistic chosen for its immunity to a spatially uniform source does not survive its own background control, an equal-area annulus of each box’s surroundings aligning at least as well as the object in four of them, an alignment that is ego-motion, which this data does not separate from object timing. The narrow-exposure contrast is one sequence, so width and illumination move together in it; a denoiser ablation is not run.

On the predictor side, everything is one dataset and 5 checkpoints of two architectures sharing a repository, and occlusion reports a sensitivity that equals a linear weight only for a predictor linear in its bins. The support measurement weights windows by a difference norm and neither fill is the training distribution, so its shares are a ranking and its centroid is horizon-dependent. The output-time measurement is a null bounded by its precision: it excludes an offset as large as the newest window’s centroid at 3.65 standard errors, says nothing about a few milliseconds, and assumes constant velocity over the label neighbourhood; the specification was chosen by a coefficient orthogonal to v^* , though the step returning it to null also moved τ .

6 Conclusion

A nominal benchmark timestamp is not a temporal specification. In the released detector studied here, the evidence supporting a prediction extends before the instant its output represents: the newest window is centered 23.81 ms before the label instant, within 1.24 ms across 5 checkpoints, and the output is bounded away from that window at 3.65 standard errors under geometric controls. Average precision is nearly invariant over that separation, 0.04 points and no reordering, so an unchanged score does not establish temporal alignment. DSEC supplies the label-side case: exposure-conditioned temporal structure bounds what event timing inside one exposure can be attributed to object timing. Predictor support, output time and label support are three quantities to declare or measure separately when temporal interpretation matters.

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